



STM Exercise 8

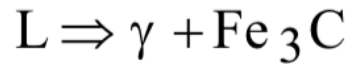
Phase Diagrams (Fe-C)

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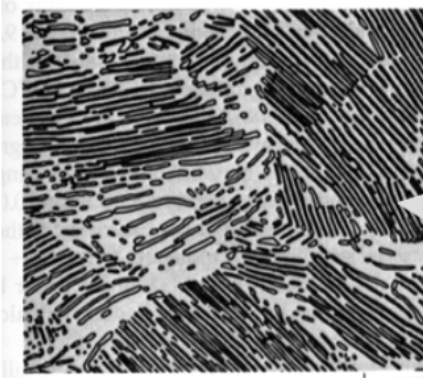
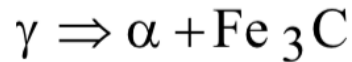
IRON-CARBON (Fe-C) PHASE DIAGRAM

- 2 important points

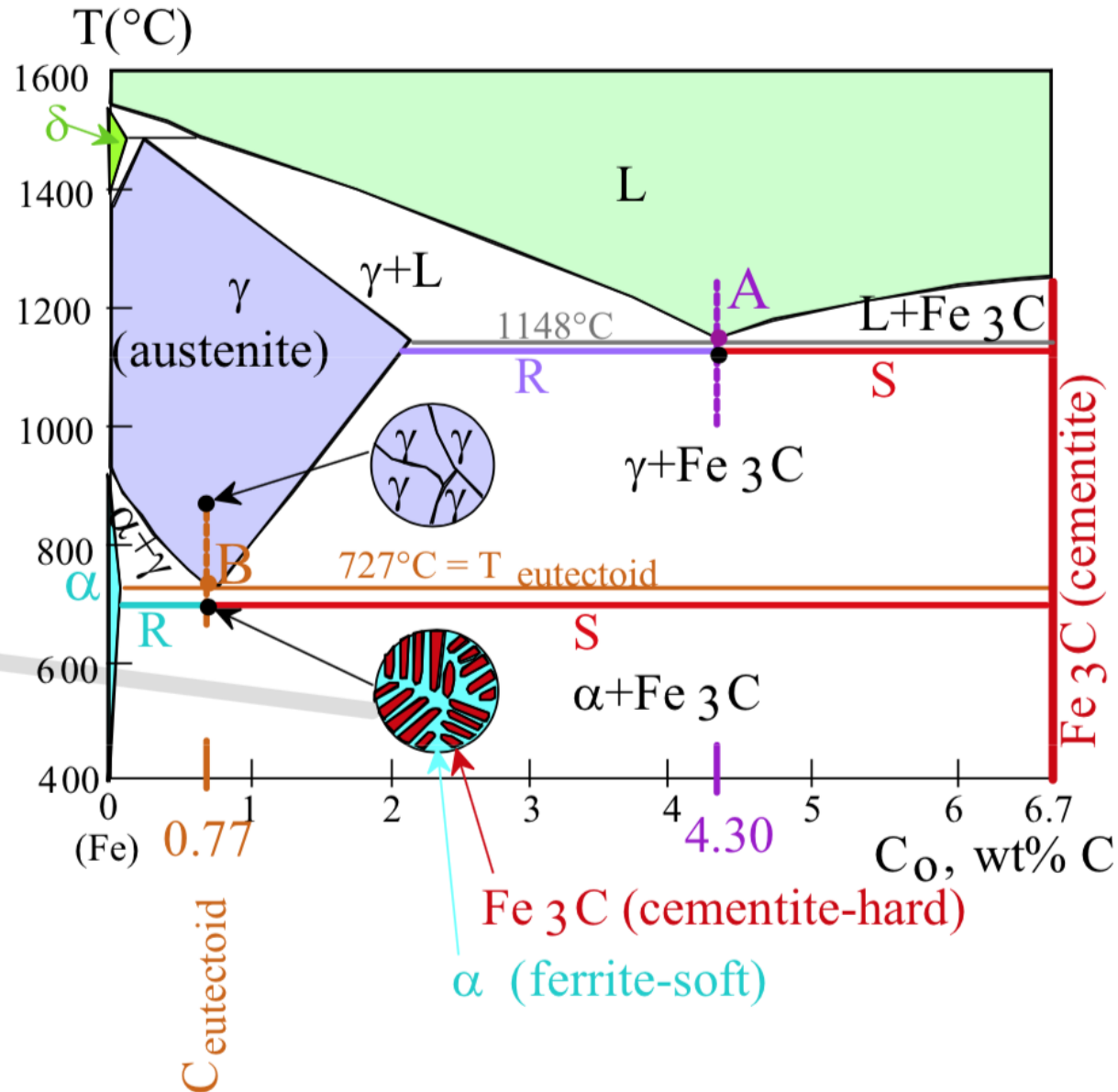
-Eutectic (A):

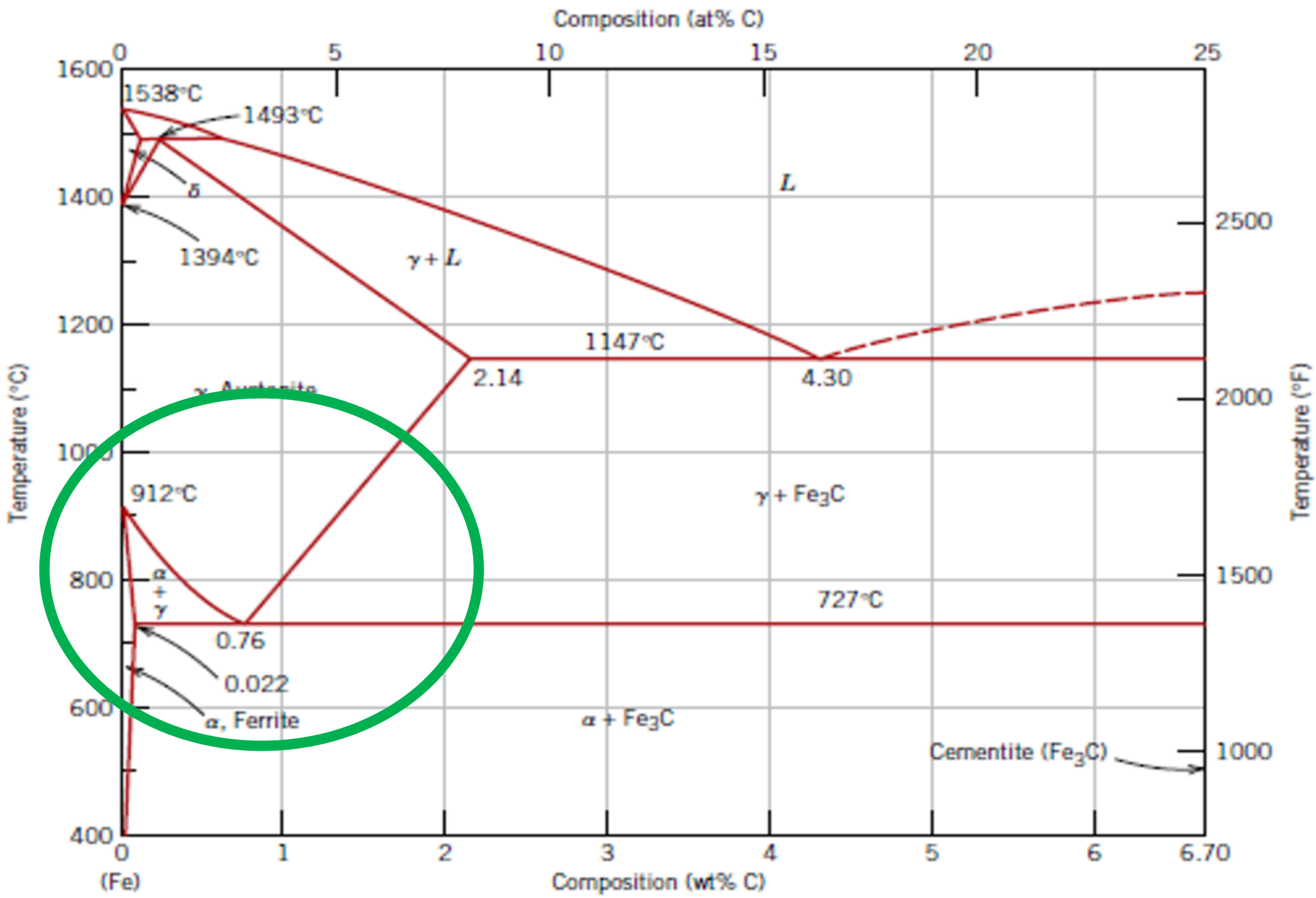


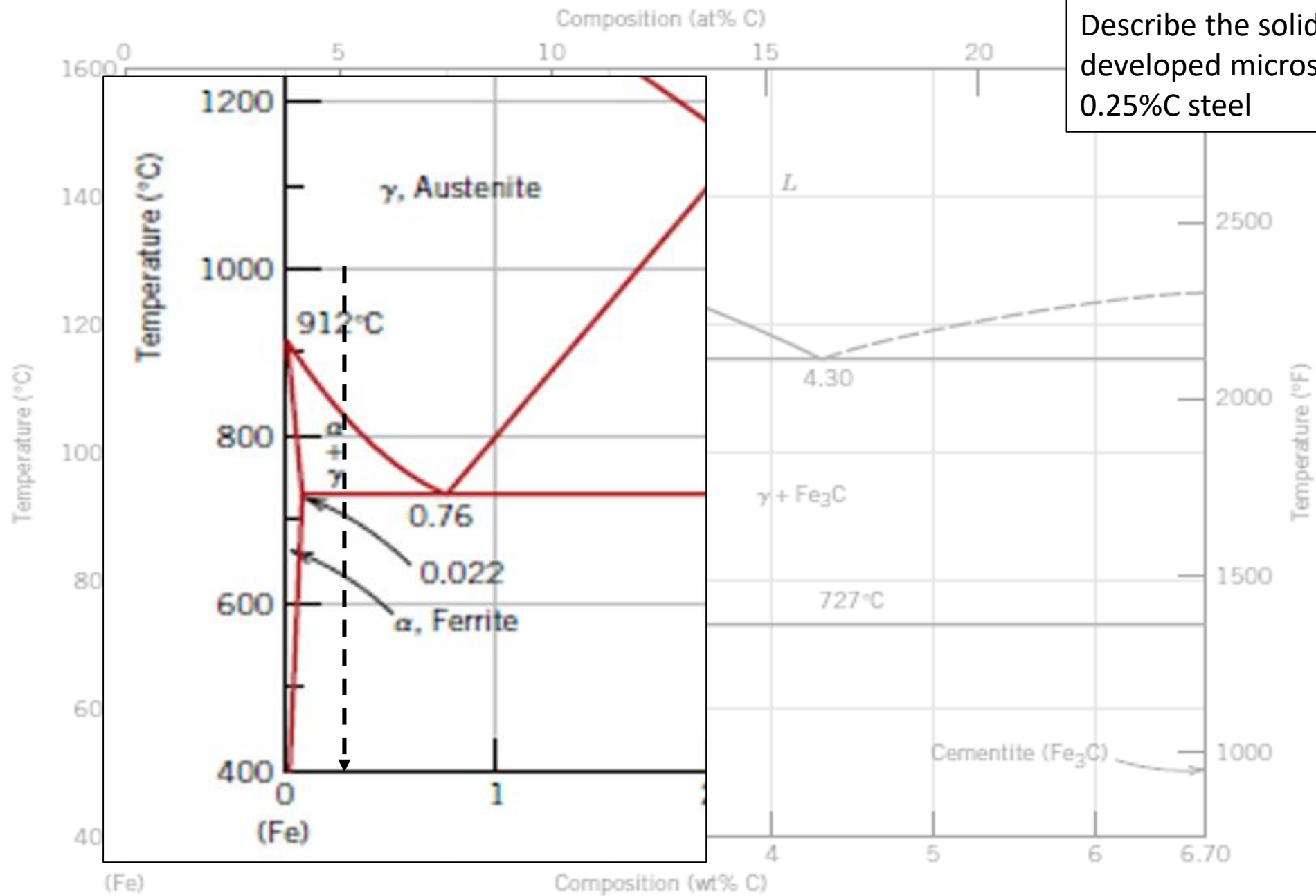
-Eutectoid (B):



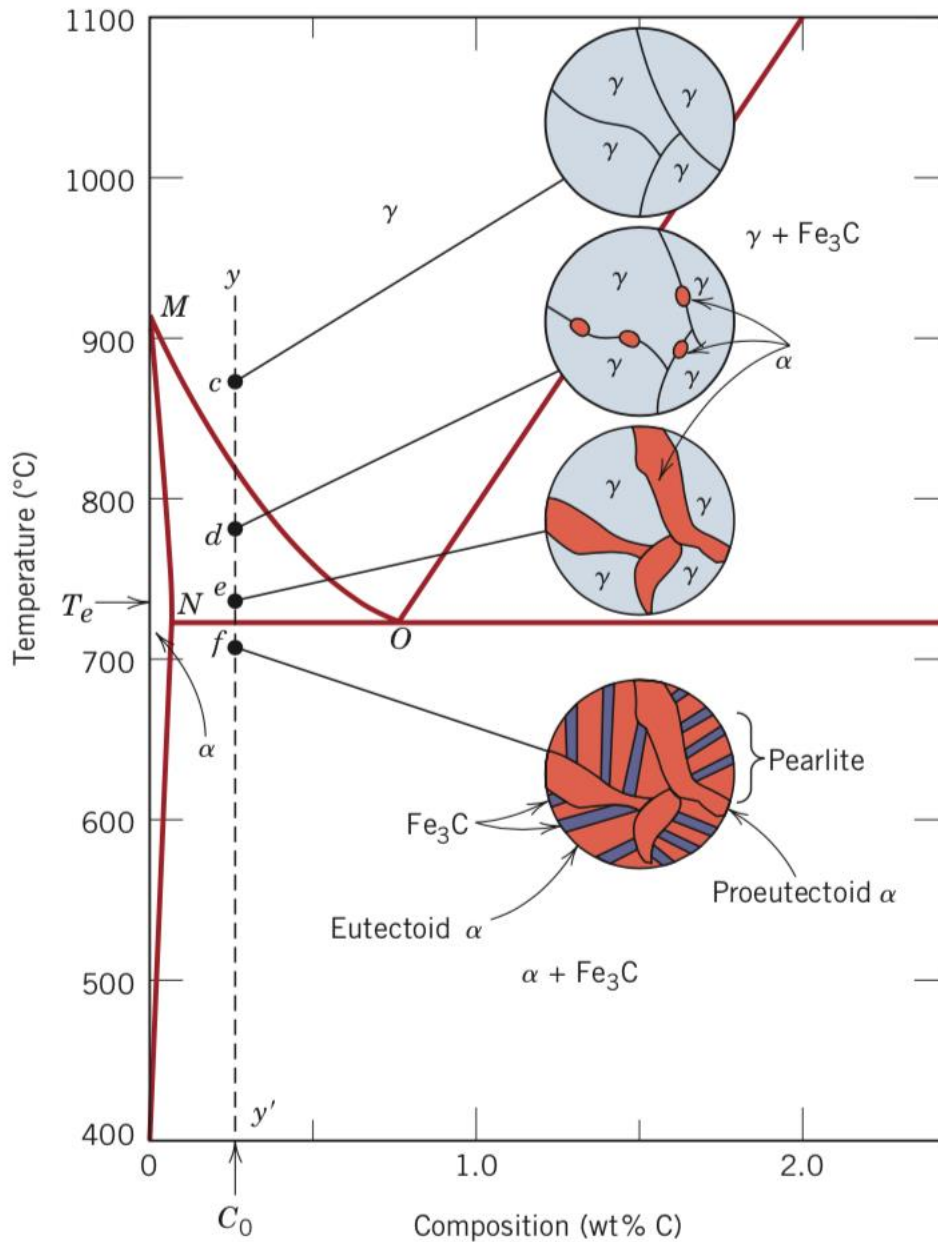
Result: Pearlite = alternating layers of α and Fe_3C phases.







Describe the solidification (evidencing developed microstructures) of a 0.25%C steel



IPO-EUTECTOIDIC STEEL

c) Solid austenite (γ)

d)-e) Ferrite (α) nucleation and growth

Eutectoid temperature: complete solidification

$\gamma \rightarrow \alpha + \text{Fe}_3\text{C}$

f) $\alpha + \text{Fe}_3\text{C}$ (pearlite) around α (proeutectoid ferrite)

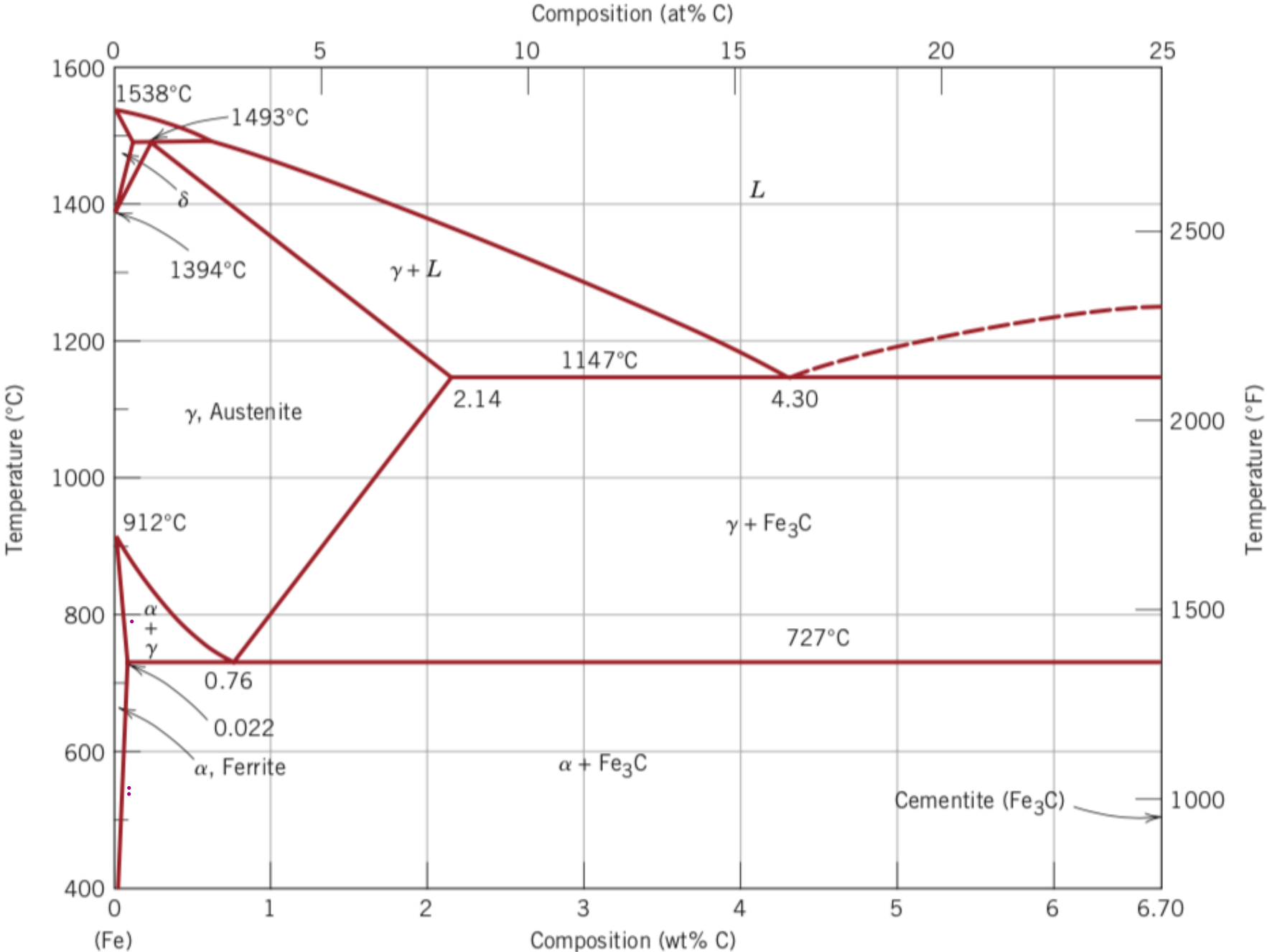
α Proeutectoid alpha(ferrite)

Eutectoid reaction: Austenite $\rightarrow$ Ferrite + Cementite

Pearlite

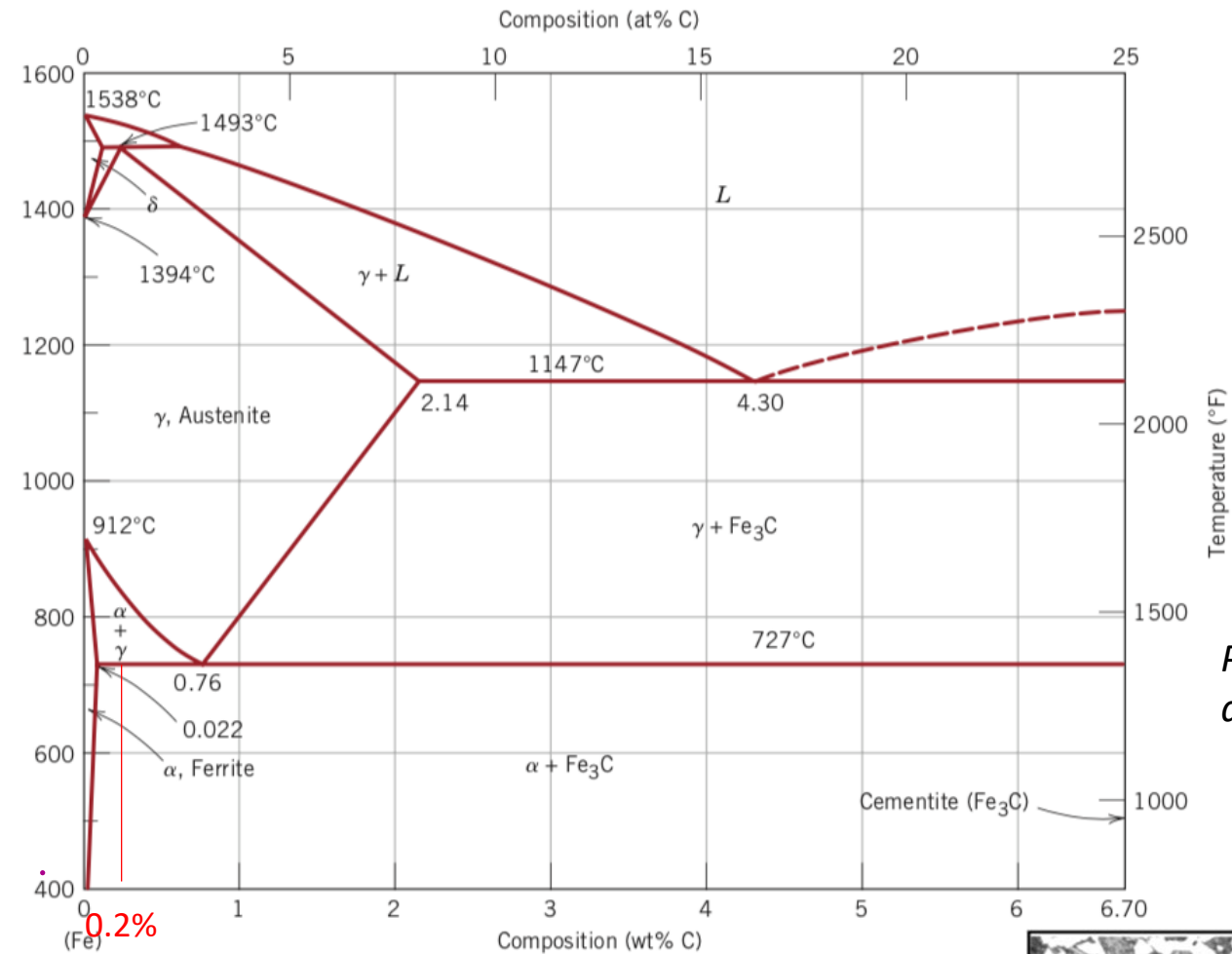
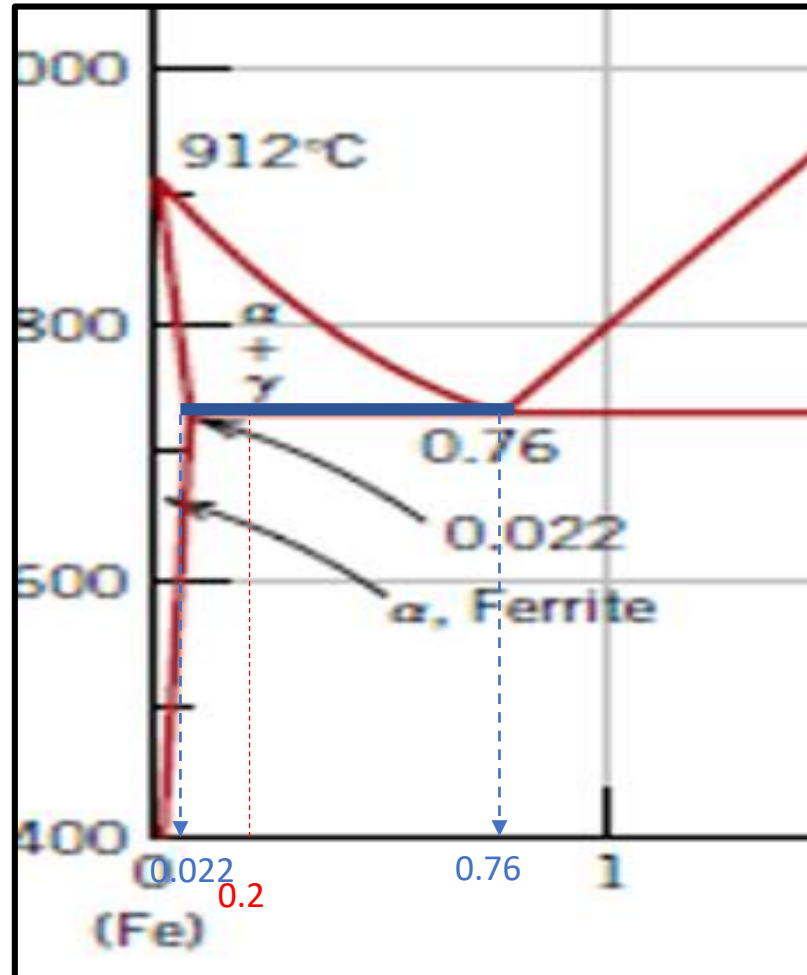
Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of primary (proeutectoid) ferrite for a 0,2 % C steel



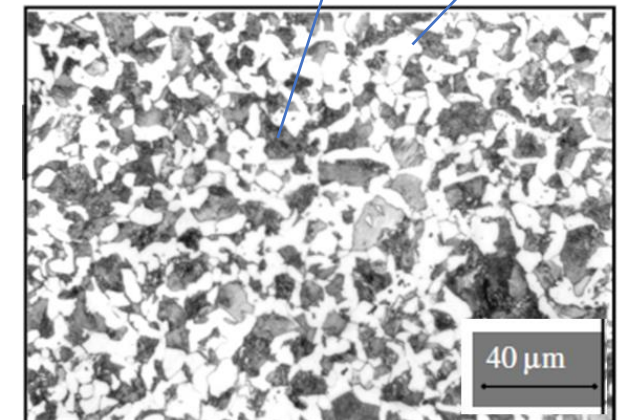
Use the Fe-Fe₃C phase diagram to calculate:
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-the percentage of primary (pro-eutectoid) ferrite for a 0,2 % C steel



Pearlite (lamellar dark grains)

Primary ferrite (white grains)

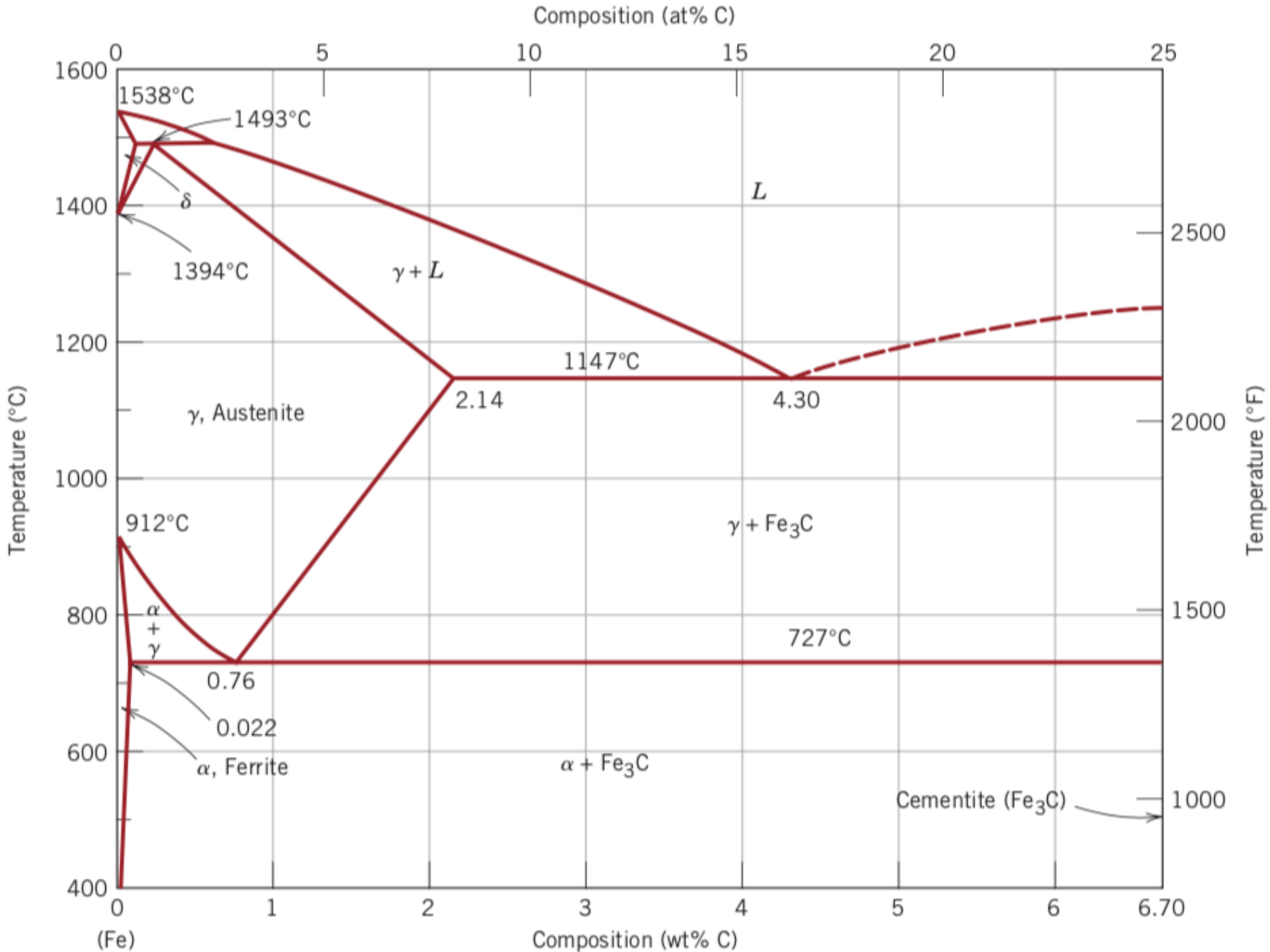


Lever rule on the eutectoidic tie line (blue line in figure):

$$\% \alpha = (0.76 - 0.2) / (0.76 - 0.022) * 100 = 76\%$$

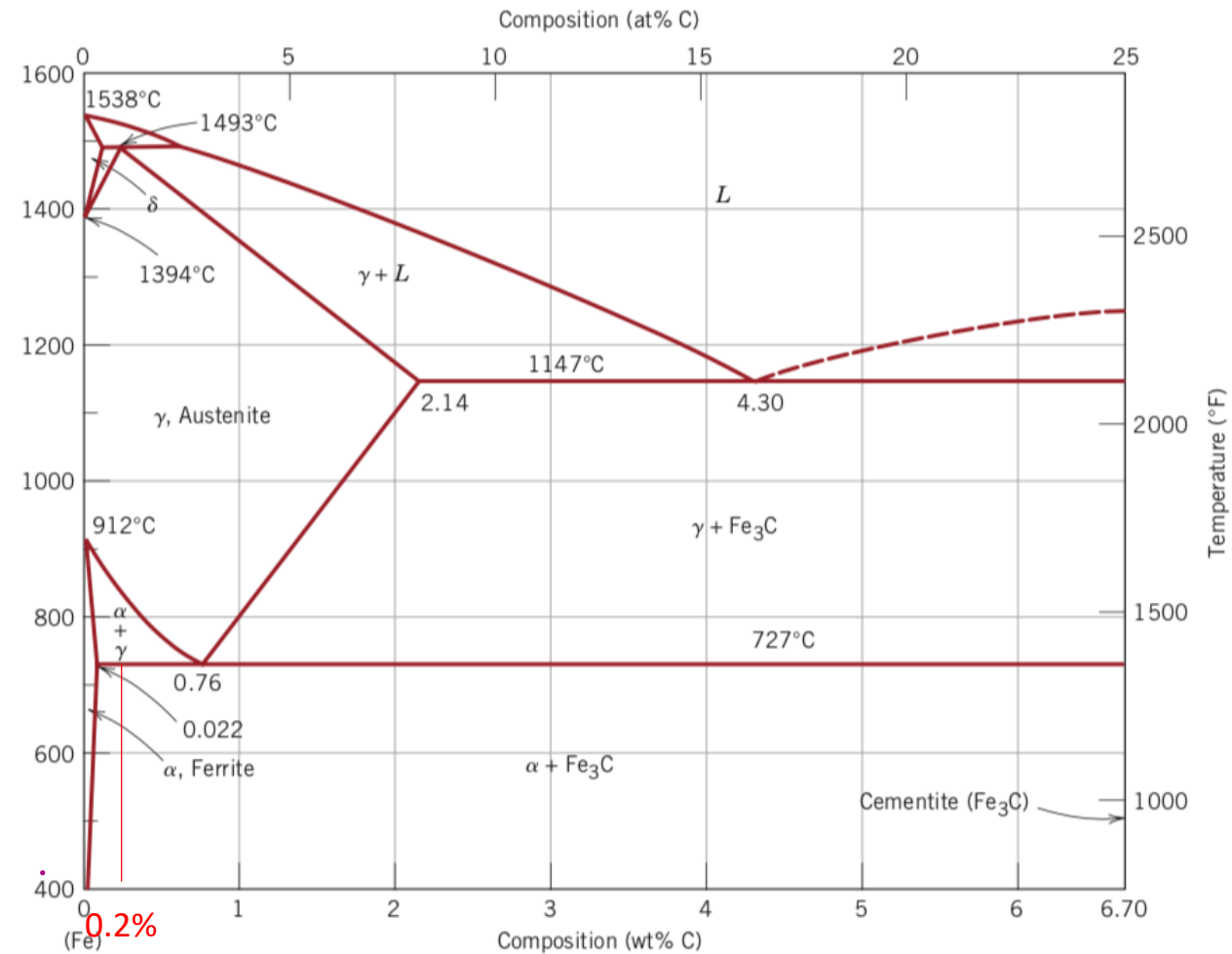
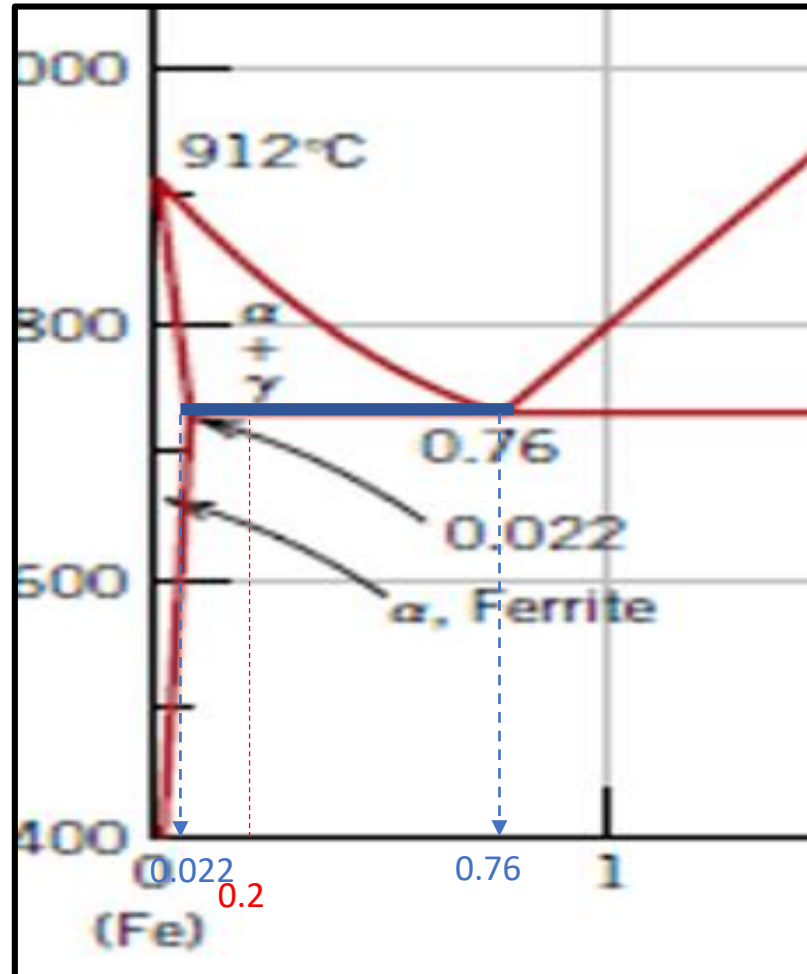
Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel



Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of pearlite for a 0,2 % C steel



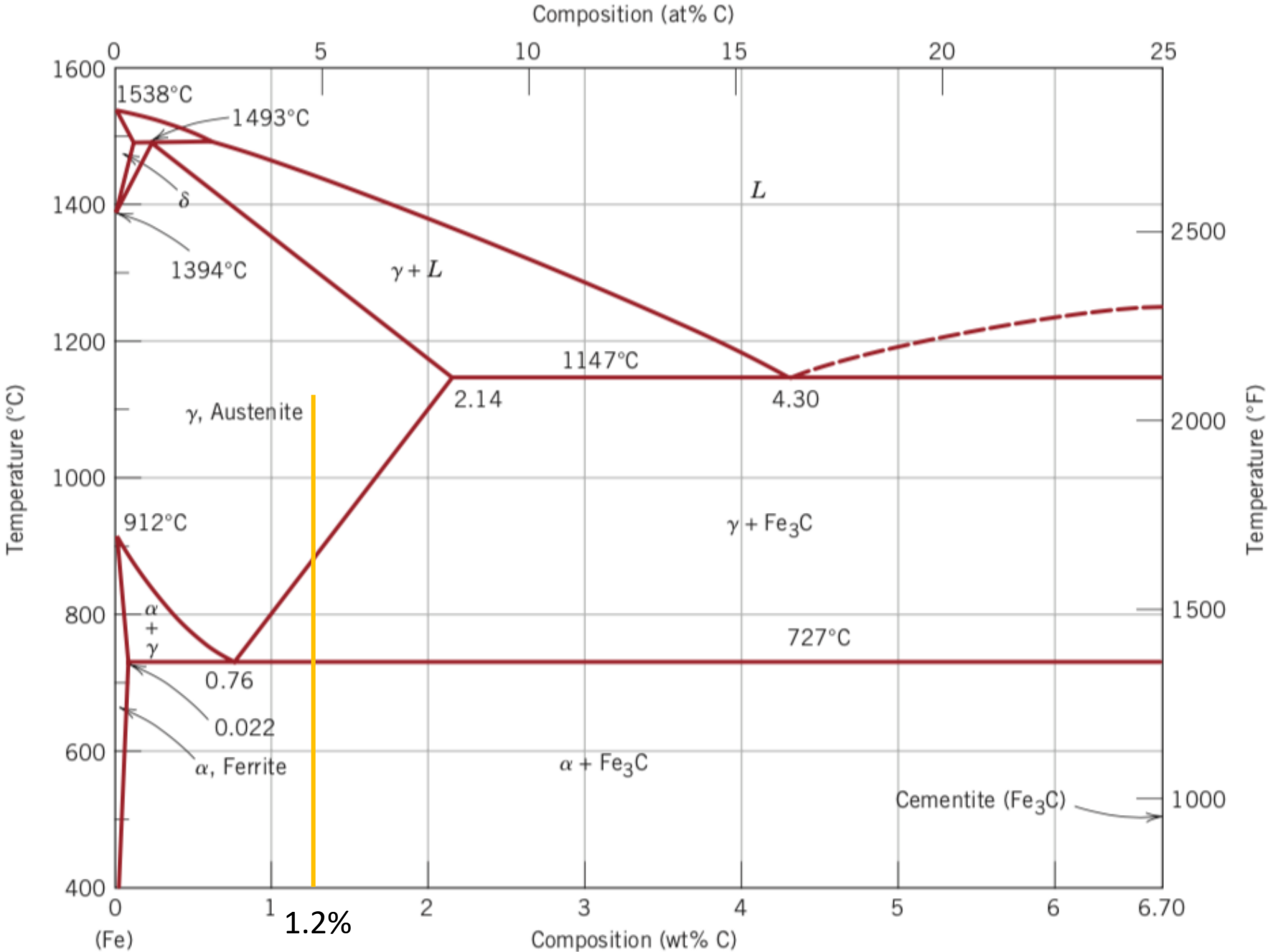
Lever rule on the eutectoid tie line (blue line in figure):

$$\% \gamma = (0.2 - 0.022) / (0.76 - 0.022) * 100 = 24\%$$

The % of pearlite is equal to the one of austenite just before the autectoid

Use the Fe-Fe₃C phase diagram to calculate:

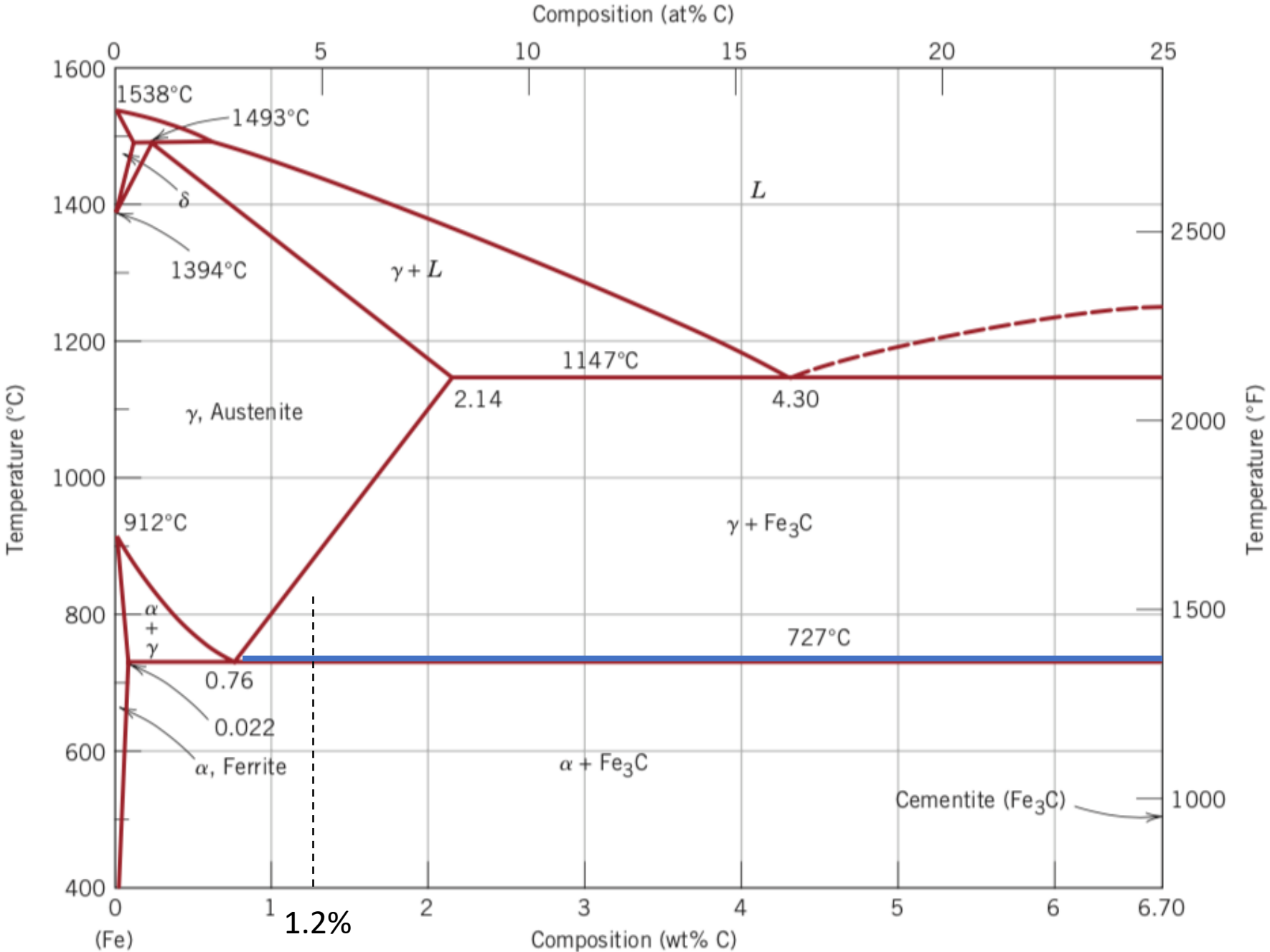
-the percentage of pearlite for a 1,2 % C steel



Use the Fe-Fe₃C phase diagram to calculate:

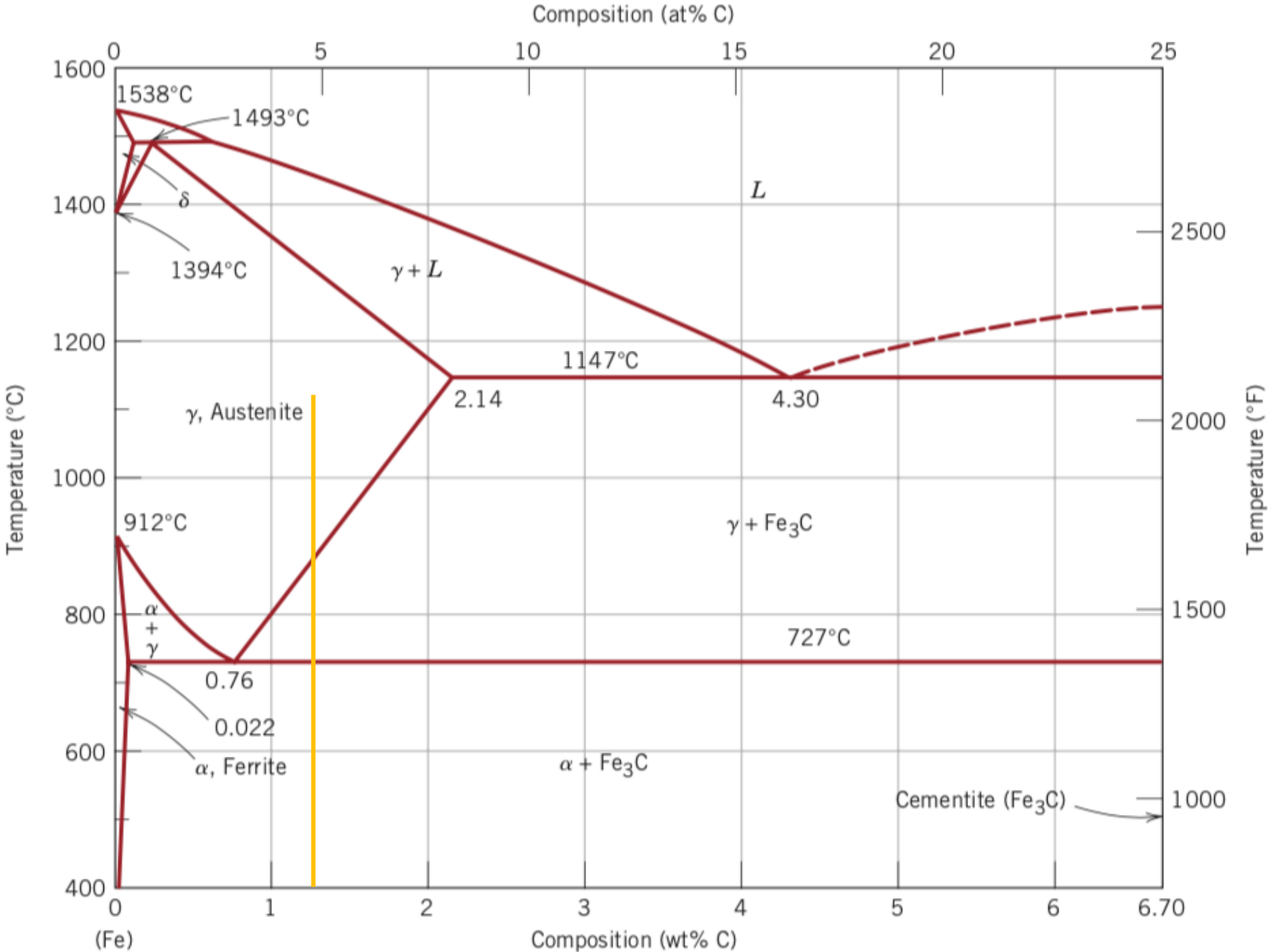
-the percentage of pearlite for a 1,2 % C steel

During eutectoidic transformation all the austenite (γ) is transformed in α +Fe₃C pearlitic structure.
So, the % of pearlite is equal to the % of γ at the beginning of the eutectoidic transformation
 $\% \gamma (723^{\circ}\text{C}) = (6.70 - 1.2) / (6.70 - 0.76) * 100 = 92.5\%$



Use the Fe-Fe₃C phase diagram to calculate:

-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel



Use the Fe-Fe₃C phase diagram to calculate:
to calculate:

-the percentage of primary (pro-eutectoid) cementite for a 1,2 % C steel

Lever rule at 727°C
 $\%Fe_3C \text{ (primary)} = \frac{(1.2 - 0.76)}{(6.70 - 0.76)} \times 100 = 7.4\%$

